AISHIFA HEALTH CLUSTER ORCHESTRATION ENGINE

Enterprise Sovereign Architecture & Clinical Decision Support System (CDSS)

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1. Executive Summary & Professional Background

AiShifa Argon 4 Pro is an enterprise-grade, 7-agent sovereign clinical grid engineered to automate hospital workflows, optimize emergency response times, and enforce rigorous regulatory compliance under Saudi Arabian health frameworks (MOH, SFDA SaMD, SDAIA PDPL, NPHIES, and CCHI). Backed by **15+ years of clinical healthcare experience** and comprehensive management and AI credentials (including HP Leadership Skills, Great Learning Healthcare Management & AI Applications, UniAthena Healthcare Management, and Barthel Index Assessment), the platform delivers verifiable efficiency gains—projecting up to **SAR 9.8M in annual operational savings per 500-bed hospital site**.

2. Core System Metrics & Professional Certifications

- 15+ Years Clinical Healthcare & Administrative Experience
- 11,622 Enterprise Verification Points | 41 Verified GCP Badges | 12 Gear Badges
- 17 SCFHS Verified Credentials & 45 Accredited CPD Credits
- **Executive & Technical Certifications:** HP Leadership Skills, Great Learning Healthcare Management, Great Learning AI Applications in Healthcare, UniAthena Basics of Healthcare Management, and Barthel Index Assessment.

3. The 7 Specialized Autonomous Agents (Google Cloud ADK & MCP)

Powered by an online **Google Cloud ADK Event Bus** and Model Context Protocol (MCP), the architecture deploys 7 specialized autonomous agents coordinating clinical care:

1. **Emergency Triage Agent:** Inbound Red Crescent EMS telemetry tracking, STEMI/Stroke standby alerts, and CTAS acuity scoring.
2. **Bilingual Clinical & Regulatory Decision Matrix:** Synchronous Arabic/English shift handovers, automated ICD-10/11 cross-matching, and prescription/NPHIES authorization validation.
3. **NPHIES Clean Tariff & Scrubber Agent:** Automated pre-authorization scrubbing, ACHI code mapping, and anti-fraud claim validation.
4. **Resuscitation Beds Live Board:** Real-time telemetry tracking of resuscitation bays, Cath Lab readiness, and active patient ED monitoring.
5. **D2B Metrics & Analytics:** Real-time tracking of critical clinical benchmarks (Door-to-Doctor ~14.8 mins; Door-to-Balloon PCI ~46 mins vs. 90-min target).
6. **Biometric Governance & Admin Agent:** Memory-hard sovereign isolation ensuring zero raw biometric exposure under SDAIA PDPL guardrails.
7. **Cluster Command & Analytics Lead:** Executive-level operational visibility across regional health clusters.

4. Regulatory & Sovereign Compliance Standards

- **Data Residency:** In-kingdom deployment (*me-central2* Dammam/Riyadh KMS).

• **Regulatory Frameworks:** Fully aligned with SDAIA PDPL, SFDA Software as a Medical Device (SaMD), Saudi MOH, NPHIES, and CCHI guardrails under a strict *Doctor-in-the-Loop* clinical decision governance protocol.

• **Interoperability Standard:** HL7 FHIR R4 v4.0.1.

5. Projected Economic & Operational Impact

• **Revenue Cycle Management (RCM):** Automated NPHIES tariff scrubbing reduces initial claim rejections and administrative audit friction (~3.8M SAR/year).

• **Emergency & Cath Lab Optimization:** Streamlined D2B (46 mins) and D2D (14.8 mins) workflows accelerate patient throughput and bed turnover (~2.6M SAR/year).

• **Clinical Documentation Efficiency:** Bilingual automated handovers and discharge summaries save 8–9 minutes per patient case for clinical staff (~1.9M SAR/year).

• **Compliance & Risk Mitigation:** Automated anti-fraud auditing and zero-biometric exposure protect against regulatory penalties (~1.5M SAR/year).

• **Total Estimated Annual Savings: SAR 9.8M per hospital / health cluster hub.**